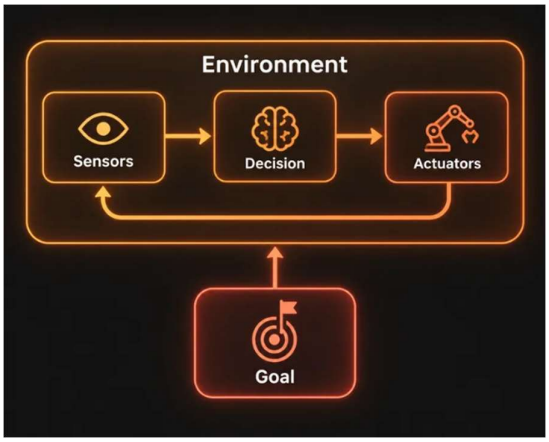
What is an Intelligent Agent?

An **Intelligent Agent** is a system that can **observe its environment, make decisions, and perform actions** to achieve a particular goal.

A self-driving car is an intelligent agent.

- **Sensors:** Camera, GPS, radar
- **Environment:** Roads, traffic, pedestrians
- **Decision:** Stop, turn, accelerate, slow down
- **Actuators:** Steering, brakes, accelerator
- **Goal:** Safely reach the destination



How Does an Intelligent Agent Work?

The basic working can be understood in five steps:

Step 1: Perceive- The agent receives information from the environment.

Step 2: Process- It analyzes the information.

Step 3: Decide- It selects an appropriate action.

Step 4: Act- It performs the selected action.

Step 5: Learn- Some intelligent agents learn from previous experiences and improve their decisions.

Example: Robot Vacuum Cleaner



What is Rationality?

Rationality means **choosing the best possible action based on the information available, with the goal of achieving the desired result.**

Rational Agent

A rational agent is an agent that chooses the action expected to produce the best outcome according to its goals and available information.

Example - A self-driving car:

Perceives:

- Traffic lights
- Other vehicles
- Pedestrians
- Road signs

Decides:

- Stop
- Turn
- Accelerate
- Brake

Acts:

- Applies brakes when a pedestrian is detected.

If stopping helps avoid an accident, **stopping is the rational action.**

Life cycle of a Rational agent

1.Environment

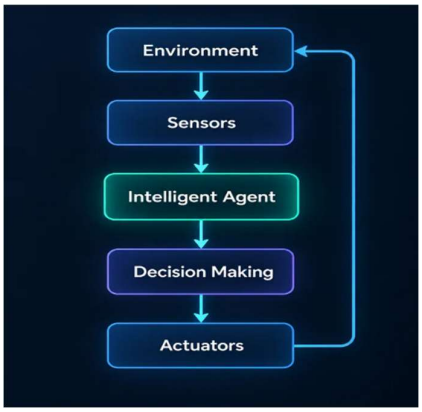
2.Sensor

3.Agent

4.Decision

5.Actuator

6.Environment



Rationality Depends on Four Things

The rational action of an agent depends on:

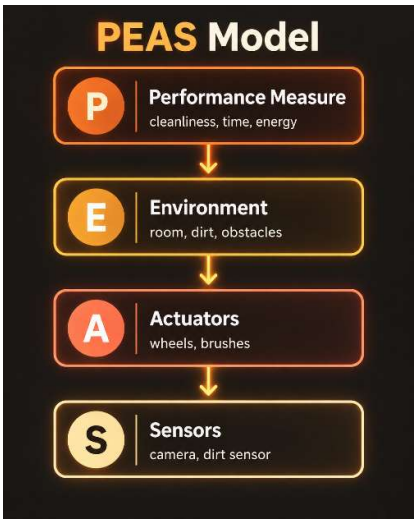
1. **Performance measure**
2. **Percept sequence**
3. **Knowledge about the environment**
4. **Available actions**

PEAS Model

PEAS is commonly used to describe the task environment of an intelligent agent.

PEAS stands for:

P – Performance Measure
E – Environment
A – Actuators
S – Sensors



PEAS	Example
Performance	Win the game
Environment	Chess board
Actuators	Move chess pieces
Sensors	Board/game state

Types of Intelligent Agents

There are **five major types of agents** commonly discussed in AI.

1. Simple Reflex Agent

It makes decisions based only on the **current percept**.

Rule:

IF condition → THEN action

Example:

Automatic door:

IF person detected

THEN open door

Advantages

- Simple
- Fast
- Easy to implement

Disadvantages

- Cannot remember the past

- Cannot handle complex situations
- Works poorly when information is incomplete

2.Model-Based Reflex Agent

A model-based agent maintains some **internal information about the world**.

It can remember previous situations.

Example

A robot knows:

"I was in Room A, then I moved to Room B."

It uses this internal state to make better decisions.

Advantage

It can work in environments where the complete current state cannot be directly observed.

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3. Goal-Based Agent

A goal-based agent takes actions to **achieve a specific goal**.

Example: Google Maps

Goal:

Reach Delhi from Lucknow.

The system considers different routes and selects actions that help reach the destination.

Current Location

↓

Goal

↓

Find Possible Routes

↓

Select Route

↓

Reach Destination

Advantage

It can consider different possible actions before making a decision.

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4. Utility-Based Agent

A utility-based agent doesn't just ask:

"Can I achieve the goal?"

It also asks:

"Which choice gives me the best result?"

Example

Suppose there are three routes:

- Route A → 3 hours
- Route B → 2 hours
- Route C → 2.5 hours but safer

A utility-based agent may consider:

- Time
- Safety
- Cost
- Comfort

and select an action based on the overall utility.

***(Utility** represents how desirable or beneficial a particular outcome is.)

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5. Learning Agent

A learning agent can **learn from experience and improve its performance over time**.

Example: Recommendation System

Suppose you frequently watch Java programming videos.

The system learns your behavior and may recommend:

"Spring Boot Tutorial"

because it has learned your interests.
